

COURTNEY CARREIRA

(she/her)

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EDUCATION

Ph.D. Candidate

September 2023 - Present

Department of Astronomy & Astrophysics, The University of California, Santa Cruz

Santa Cruz, CA

Advisor: Professor Brant Robertson

M.S. Astronomy and Astrophysics

June 2025

Department of Astronomy & Astrophysics, The University of California, Santa Cruz

Santa Cruz, CA

B.S. Physics

Graduated with General Honors, May 2023

Department of Physics and Astronomy, Johns Hopkins University

Baltimore, MD

Minor in Applied Mathematics and Statistics

RESEARCH EXPERIENCE

Graduate Student Researcher

January 2024 - Present

Department of Astronomy & Astrophysics, The University of California, Santa Cruz

Santa Cruz, CA

- As a member of the *JWST* Advanced Deep Extragalactic Survey (JADES) collaboration, I am working to understand how morphology traces galaxy evolution across cosmic time. I organize the Morphology Working Group within the collaboration.
- Created catalogs of morphological parameters by using Bayesian techniques to robustly perform model-fitting of Sérsic profiles to $> 10^5$ galaxies in GOODS-N and GOODS-S, resulting in over 3 million Sérsic profiles computed and shared with the community as part of JADES Data Release 5.

NSF REU Intern and NRAO SOS Researcher

June 2022 - September 2023

Smithsonian Astrophysical Observatory

Cambridge, MA

- Using observations of atomic and molecular gas emissions in M33 to analyze the effect of proximity from the southeastern spiral arm in the formation of molecular clouds; advised by Dr. Eric Koch and Dr. Sarah Jeffreson, within Professor Alyssa Goodman's research group.
- Ongoing work resulted in a successful NRAO Student Observing Support award to obtain observations that resolve the filamentary morphology of molecular clouds across M33. Co-PI: Eric Koch, Title: *Linking the Resolved Filamentary Molecular ISM to Massive Star Formation across M33*.

Undergraduate Researcher

May 2021 - May 2022

Department of Physics and Astronomy, Johns Hopkins University

Baltimore, MD

- Collected photometric and spectroscopic data for a large set of low-metallicity stellar objects, believed to host transiting exoplanets; advised by Professor Kevin Schlaufman.
- Utilized Python coding and packages to numerically analyze the stellar objects of interest.

Undergraduate Research Intern

May 2021 - January 2022

The Johns Hopkins University Applied Physics Laboratory

Laurel, MD

- Performed correlation analysis of simulated gamma-ray and UVOIR emissions from Type Ia supernovae, and assisted with scientific validation for mission proposal; advised by Dr. Richard S. Miller.

CIRCUIT Intern

April 2020 - May 2021

The Johns Hopkins University Applied Physics Laboratory

Laurel, MD

- Analyzed Monte Carlo simulations of volatile transport across the lunar surface, specifically looking at water and carbon dioxide; advised by Dr. Parvathy Prem and others.

PUBLICATIONS

First Author

1. Carreira, C., et al. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Catalogs of inferred morphological properties of galaxies from JWST/NIRCam imaging in GOODS-N and GOODS-S. arXiv e-prints, arXiv:2601.15957.

Co-Author

12. Zhu, Y., Rieke, M., Carreira, C., et al. (2026). PhotoIFU: NIRCam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies. arXiv e-prints, arXiv:2607.23020.
11. Duan, Q., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: stellar population catalogue for galaxies in GOODS-N and GOODS-S. arXiv e-prints, arXiv:2605.21599.
10. Rinaldi, P., et al. incl. Carreira, C. (2026). The Way We Tally Becomes the Tale: the Impact of Selection Strategies on the Inferred Evolution of Little Red Dots Across Cosmic Time. arXiv e-prints, arXiv:2604.07138.
9. Rinaldi, P., et al. incl. Carreira, C. (2026). Filling the Gap in Cluster Evolution: JWST's Glimpse into a Young, Star-Forming Cluster at Cosmic Noon. arXiv e-prints, arXiv:2602.24162.
8. Robertson, B., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometric Catalog. arXiv e-prints, arXiv:2601.15956.
7. Johnson, B., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: NIRCam Imaging in GOODS-S and GOODS-N. arXiv e-prints, arXiv:2601.15954.
6. Cameron, A., Carreira, C., et al. (2026). JADES: Evolution of nitrogen abundances in star-forming galaxies from $z \sim 1.5 - 7$. MNRAS, stag1575. DOI: [10.1093/mnras/stag1575](https://doi.org/10.1093/mnras/stag1575)
5. Hainline, K., et al. incl. Carreira, C. (2026). JWST Advanced Deep Extragalactic Survey (JADES) Data Release 5: Photometrically Selected Galaxy Candidates at $z > 8$. ApJ, 1004(2), 161. DOI: [10.3847/1538-4357/ae710a](https://doi.org/10.3847/1538-4357/ae710a)
4. Zhu, Y., et al. incl. Carreira, C. (2026). Clump-like Structures in High-Redshift Galaxies: Mass Scaling and Radial Trends from JADES. The Astrophysical Journal, 1000(2), 303. DOI: [10.3847/1538-4357/ae50f1](https://doi.org/10.3847/1538-4357/ae50f1)
3. Danhaive, A., et al. incl. Carreira, C. (2026). Beyond the stars: Linking $H\alpha$ sizes, kinematics, and star formation in galaxies at $z \approx 4 - 6$ with JWST grism surveys and geko. MNRAS, 547(4), stag437. DOI: [10.1093/mnras/stag437](https://doi.org/10.1093/mnras/stag437)
2. Shields, T., et al. incl. Carreira, C. (2026). JADES: low surface brightness galaxies at $0.4 < z < 0.8$ in GOODS-S. MNRAS, 546(4), stag202. DOI: [10.1093/mnras/stag202](https://doi.org/10.1093/mnras/stag202)
1. Robertson, B., et al. incl. Carreira, C. (2024). Earliest Galaxies in the JADES Origins Field: Luminosity Function and Cosmic Star Formation Rate Density 300 Myr after the Big Bang. ApJ, 970(1), 31. DOI: [10.3847/1538-4357/ad463d](https://doi.org/10.3847/1538-4357/ad463d)

PRESENTATIONS

Oral Presentations

Characterizing the morphological evolution of high-redshift galaxies with JADES <i>Santa Cruz Galaxy Workshop 2026</i>	August 2026 <i>Santa Cruz, CA</i>
Characterizing the morphological evolution of high-redshift galaxies with JADES <i>BeyondHubble: Revisiting the Hubble Sequence Across Cosmic Time</i>	April 2026 <i>KICC, Cambridge, UK</i>
Science Review: Galaxy Morphology & Kinematics <i>JADES Team Meeting, Madrid/Boston 2025</i>	June 2025 <i>Cambridge, MA</i>

Revealing the relationship between galaxy formation and morphology across cosmic time with JADES

UCSC Friday Lunch Astrophysics Seminar Hour (FLASH)

May 2025
Santa Cruz, CA

Do spiral arms form molecular clouds?

SAO Astronomy REU Summer Symposium

August 2022
Cambridge, MA

Poster Presentations

The Effect of Spiral Arms on Molecular Cloud Formation in M33

241st Meeting of the American Astronomical Society

January 2023
Seattle, WA

Lunar Crater Maturity Analysis in Python: Developing a Toolkit for Ejecta Analysis

5th Planetary Data Workshop & Planetary Science Informatics & Analytics

June 2021
Virtual

Lunar Crater Maturity Analysis in Python: Developing a Toolkit for Ejecta Analysis

52nd Lunar and Planetary Science Conference

March 2021
Virtual

The Effect of Isotopic Composition and Surface Residence Times on Lunar Volatile Transport

52nd Lunar and Planetary Science Conference

March 2021
Virtual

AWARDS AND RECOGNITION

ARCS Foundation Scholar

Achievement Rewards for College Scientists (ARCS) Foundation, Northern California Chapter

May 2026 - Present
San Francisco, CA

Osterbrock Leadership Program Fellow

Osterbrock Leadership Program at the University of California, Santa Cruz

December 2025 - Present
Santa Cruz, CA

- The Osterbrock Leadership Program maintains a small cohort of fellows, providing them with unique training through educational activities, mentorships with outstanding leaders, and leadership experiences conceived and executed by the fellows themselves.

Whitford Prize

Department of Astronomy & Astrophysics, The University of California, Santa Cruz

June 2025
Santa Cruz, CA

- Awarded to the second-year UCSC Astronomy & Astrophysics graduate student who, in the judgment of the faculty, attains the highest achievement in research, coursework, and teaching.

Graduate Research Fellowship Program (GRFP) Honorable Mention

National Science Foundation

April 2025
Alexandria, VA

TEACHING

Teaching Assistant for ASTR 2

Department of Astronomy & Astrophysics, The University of California, Santa Cruz

January 2024 - March 2024
Santa Cruz, CA

- Led one recitation section per week, which included a short lecture, group activities, and live demonstrations.
- Hosted office hours on a weekly basis, in collaboration with other TAs.

Teaching Assistant for General Physics I

Department of Physics and Astronomy, Johns Hopkins University

August 2022 - December 2022
Baltimore, MD

- During Active Learning sections of this course, worked closely with students as they completed a series of problems and hands-on demonstrations during their lectures.
- Hosted office hours on a weekly basis.

OUTREACH AND SERVICE

Lead Organizer, Department Journal Club

Department of Astronomy & Astrophysics, The University of California, Santa Cruz

January 2025 - Present

Santa Cruz, CA

PI, Osterbrock Rising Graduate Award Program

Osterbrock Leadership Program at the University of California, Santa Cruz

March 2024 - Present

Santa Cruz, CA

- Created award program, in collaboration with the UCSC Women in Physics and Astrophysics organization, to provide \$500 to four UCSC women and gender minority undergraduates to offset costs associated with applying to graduate school programs in physics and/or astronomy.
- Created additional award program to provide \$500 to four UCSC undergraduates who are negatively impacted by the current immigration landscape (for example, students who are undocumented or DACAmented, or whose families are) to offset costs associated with applying to graduate school programs in STEM.
- Provided mentorship and support in developing application materials to the students selected to receive this award.
- Funding generously provided by the Osterbrock Leadership Program during the 2024 and 2025 Mini-Grant cycles.

Local Organizing Committee

JADES Team Meeting, Santa Cruz 2025

January 2025

Santa Cruz, CA

Updated as of September 26, 2026.